

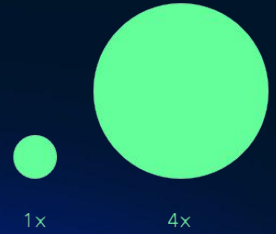
SVG

sprites

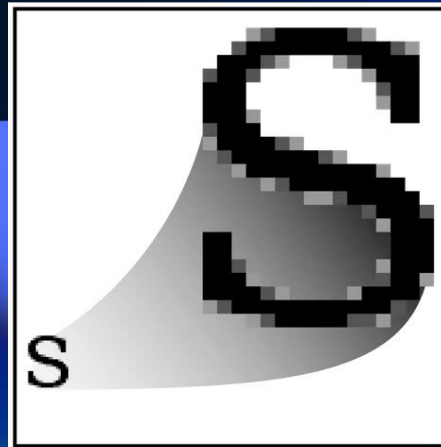


Repeat

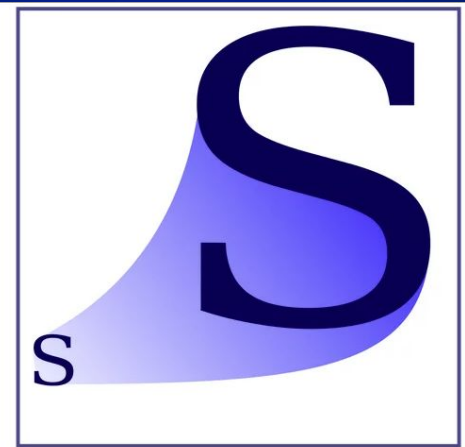
Vector images



Created using **mathematical formulas** and geometric shapes, such as **lines** and **curves**



Raster



Vector

Scalable Vector Graphics

A markup language extended from XML for describing two-dimensional vector graphics

SVG files can be created and edited with any text editor and drawing programs



Repeat

Raster vs Vector

[example →](#)

	<code>.jpeg, .png, .avif, .webp</code>	<code><svg></code>
Scales without quality loss	✗	✓
Usually has smaller size	✗	✓
How can be changed?	In special programs for graphics	In text editor or IDE
Usage for	complex images	logos, icons, decor
Parts inside can be affected through css and js	✗	✓
Can be converted to	other raster formats only	vector and raster formats

Lesson Plan

1

What is it **inside**

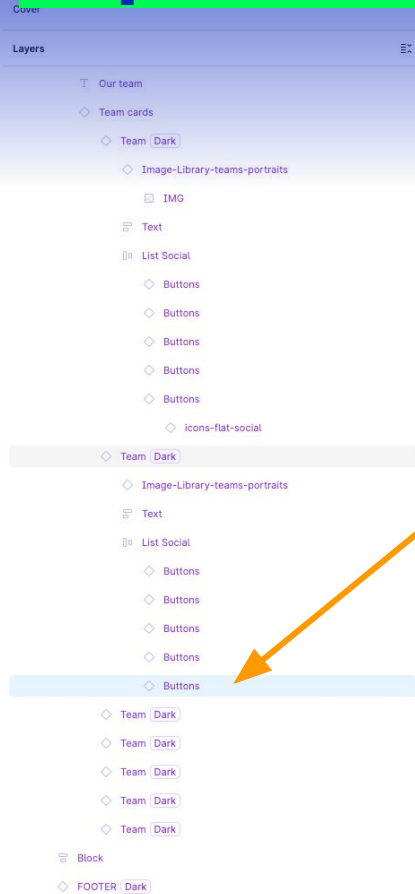
2

How to insert svg on a page

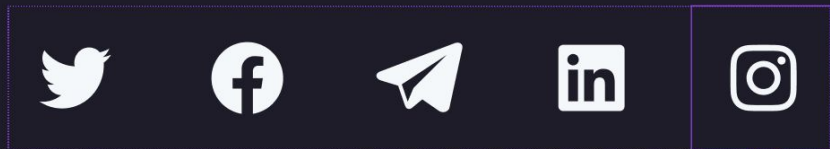
3

SVG sprite

Export from Figma



Jason Smith
Co - Founder

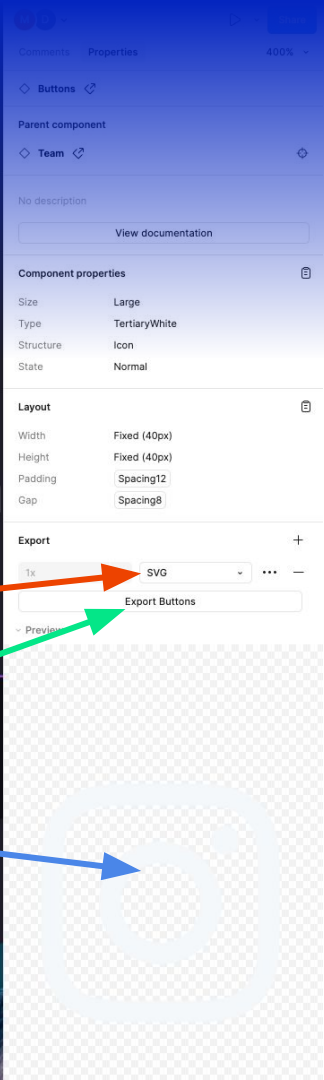


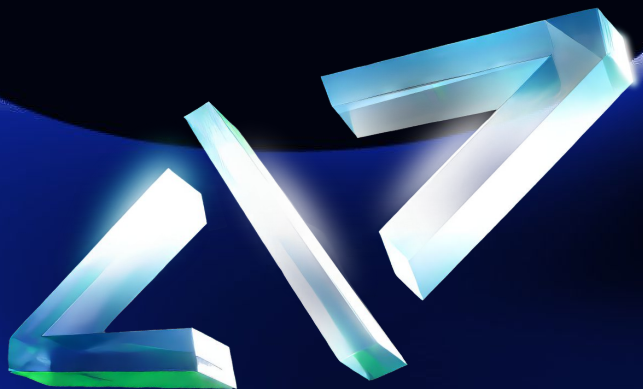
choose layer

image format

Export button

image preview





Ways to insert SVG on a page

You have an **<SVG>**

```
<svg viewBox="0 0 512 512"
xmlns="http://www.w3.org/2000/svg">
  <path d="m48 300 180 169a41 41
0 0 0 56 0l180-169c31-28 48-68
48-109v-6A143 143 0 0 0 268 84l-12
12-12A143 143 0 0 0 0 185v6c0 41
17 81 48 109z"/>
</svg>
```



Insert SVG content right into HTML



```
<a href="/" class="logo">  
  <svg viewBox="0 0 512 512">  
    <path d="m48 300 180 169a41 41 0 0 0 56 0l180-169c31-28 48-68  
48-109v-6A143 143 0 0 0 268 84l-12 12-12-12A143 143 0 0 0 0 185v6c0 41  
17 81 48 109z"/>  
  </svg>  
</a>
```



👉 don't use xmlns on svg tag

```
(xmlns="http://www.w3.org/2000/svg")
```

[example →](#)

Insert SVG content right into HTML



```
<a href="/" class="logo">  
  <svg viewBox="0 0 512 512" class="logo__icon">  
    <path d="m48 300 180 169a41 41 0 0 0 56 0l180-169c31-28 48-68  
48-109v-6A143 143 0 0 0 268 84l-12 12-12-12A143 143 0 0 0 0 185v6c0 41  
17 81 48 109z"/>  
  </svg>  
</a>
```

```
.logo__icon {  
  fill: red;  
}
```

👉 don't use xmlns on svg tag

(xmlns="http://www.w3.org/2000/svg")

[example →](#)

example →

Repeat

<picture>



```

```

👉 don't forget to use **lazy load** if image not in first viewport

Repeat

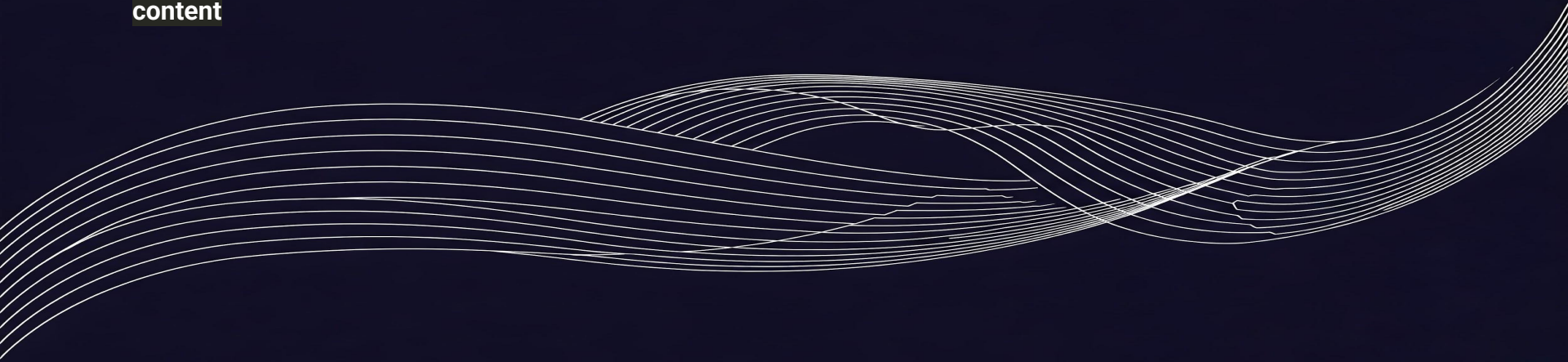
CSS background

```
background-image: url("wave-pattern.svg");
```

list-style-image

border-image

content



SVG container

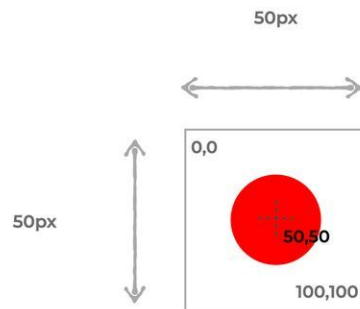


viewBox and dimensions

- Defines the aspect ratio
- Responsible for scaling

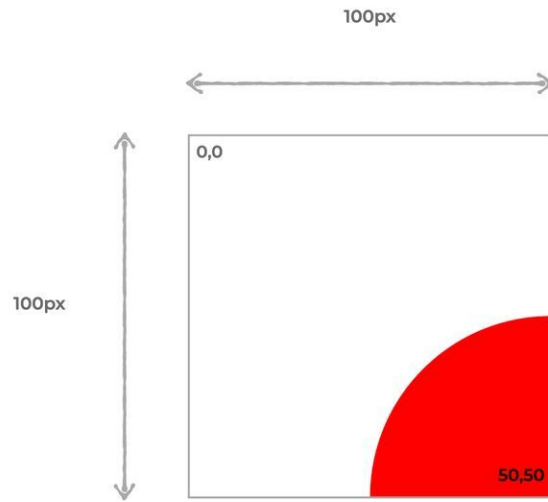
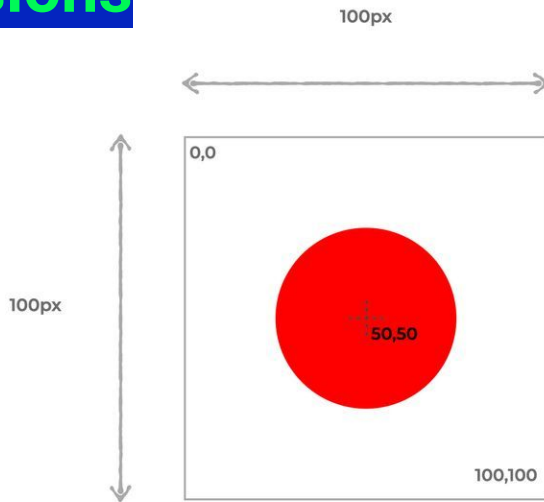
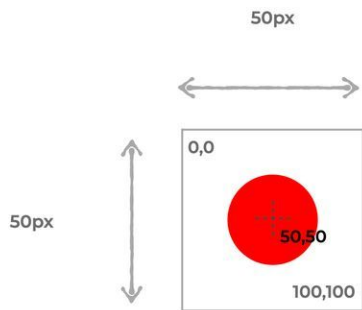
`viewBox="0 0 512 512"`

- **x** – Horizontal coordinate
- **y** – Vertical coordinate
- **width**
- **height**



```
<svg
  width="50" height="50"
  viewBox="0 0 100 100"
>
  <circle
    cx="50" cy="50"
    r="25" fill="red"
  />
</svg>
```

viewBox + dimensions



```
<svg
  width="50" height="50"
  viewBox="0 0 100 100"
>
  <circle
    cx="50" cy="50"
    r="25" fill="red"
  />
</svg>
```

```
<svg
  width="100" height="100"
  viewBox="0 0 100 100"
>
  <circle
    cx="50" cy="50"
    r="25" fill="red"
  />
</svg>
```

```
<svg
  width="100" height="100"
  viewBox="0 0 50 50"
>
  <circle
    cx="50" cy="50"
    r="25" fill="red"
  />
</svg>
```

Common attributes

- **fill** – background color
- **stroke** – border color
- **stroke-width** - border width

- Clipping, Masking and Compositing properties:
 - ['clip-path'](#)
 - ['clip-rule'](#)
 - ['mask'](#)
 - ['opacity'](#)
 - ['marker-end'](#)
 - ['marker-mid'](#)
 - ['marker-start'](#)
 - ['shape-rendering'](#)
- Filter Effects properties:
 - ['enable-background'](#)
 - ['filter'](#)
 - ['flood-color'](#)
 - ['flood-opacity'](#)
 - ['lighting-color'](#)
 - ['stroke'](#)
 - ['stroke-dasharray'](#)
 - ['stroke-dashoffset'](#)
 - ['stroke-linecap'](#)
 - ['stroke-linejoin'](#)
 - ['stroke-miterlimit'](#)
- Gradient properties:
 - ['stop-color'](#)
 - ['stop-opacity'](#)
 - ['stroke-opacity'](#)
 - ['stroke-width'](#)
 - ['text-rendering'](#)
- Interactivity properties:
 - ['pointer-events'](#)
- Color and Painting properties:
 - ['color-interpolation'](#)
 - ['color-rendering'](#)
 - ['fill'](#)
 - ['fill-opacity'](#)
 - ['fill-rule'](#)
 - ['image-rendering'](#)
 - ['marker'](#)
 - ['alignment-baseline'](#)
 - ['baseline-shift'](#)
 - ['dominant-baseline'](#)
 - ['glyph-orientation-horizontal'](#)
 - ['glyph-orientation-vertical'](#)
 - ['text-anchor'](#)
 - ['writing-mode'](#)

Insert SVG content right into HTML



```
<a href="/" class="logo">  
  <svg viewBox="0 0 512 512">  
    <path d="m48 300 180 169a41 41 0 0 0 56 0l180-169c31-28 48-68  
48-109v-6A143 143 0 0 0 268 84l-12 12-12-12A143 143 0 0 0 0 185v6c0 41  
17 81 48 109z"/>  
  </svg>  
</a>
```



👉 don't use xmlns on svg tag

```
(xmlns="http://www.w3.org/2000/svg")
```

[example →](#)

Insert SVG content right into HTML



```
<a href="/" class="logo">  
  <svg viewBox="0 0 512 512" class="logo__icon">  
    <path d="m48 300 180 169a41 41 0 0 0 56 0l180-169c31-28 48-68  
48-109v-6A143 143 0 0 0 268 84l-12 12-12-12A143 143 0 0 0 0 185v6c0 41  
17 81 48 109z"/>  
  </svg>  
</a>
```

```
.logo__icon {  
  fill: red;  
}
```

👉 don't use xmlns on svg tag

(xmlns="http://www.w3.org/2000/svg")

[example →](#)

Check Binabox design

Break time

Vector sprites



1st problem: too long loading

Each file = one HTTP request.

Too many requests = slow page load.



what if there will be only one image?

First Solution — Raster Sprite Sheets: just glue them together



Raster Sprite

Designers combined multiple icons into one large image

Only 1 image to load = 1 request!

But: hard to maintain, pixelated on retina screens, no styling via CSS.



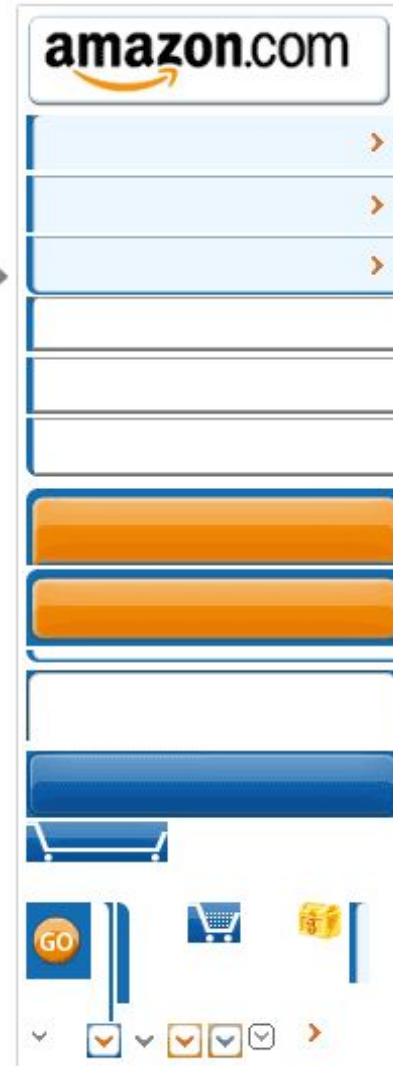
← Google



← YouTube



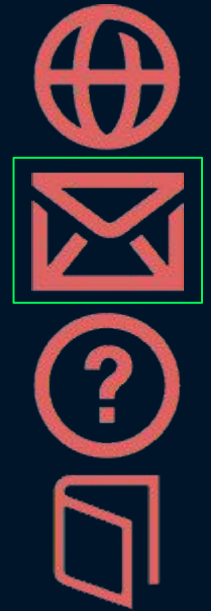
← Facebook



Amazon →

Use sprite with background

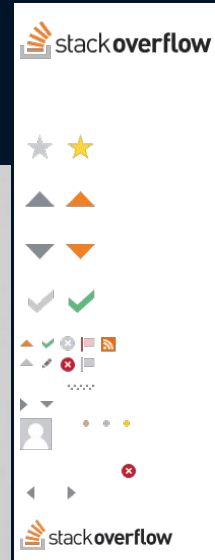
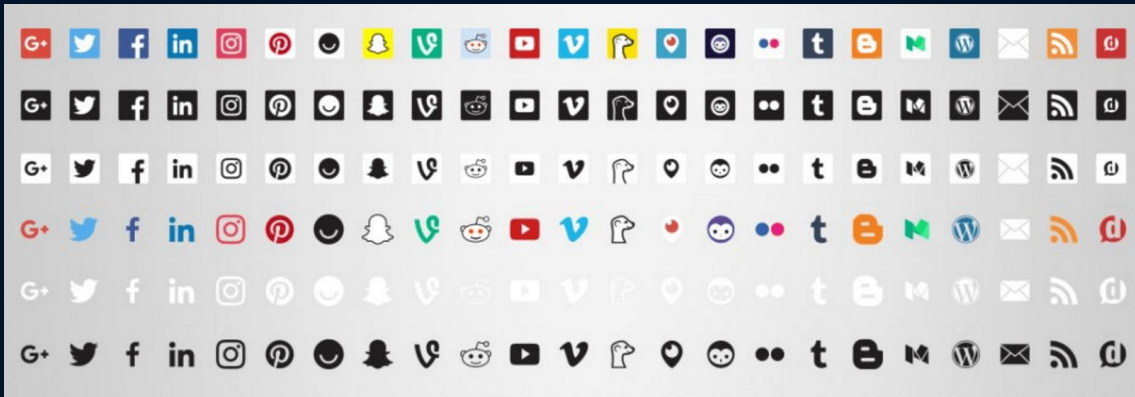
```
.sprite {  
  width: 75px; height: 75px;  
  background-image: url(sprite.png);  
}  
  
.mail { background-position: 0 -75px }
```



[example](#) →

but sprite consists of same icons

we gathered a lot of icons in our sprites,
and some icons were repeated with just changing some color



4

Then Came Icon Fonts

Huge font file
with all the icons
suitable for all projects



✗ never use

```
Filter :hov .cls +,  [icon]
element.style {
  line-height: 1;
  color: rgb(30, 48, 80);
}

:is(i[class*="fa-"], .svg-inline--fa) {
  --spacing-inline: var(--spacing-6xs);
}

.fa-solid, .fas {
  font-weight: 900;
}

.fa-home, .fa-home-alt, .fa-home-
lg-alt, .fa-house {
  --fa: "\f015";
  --fa--fa: "\f015\f015";
}

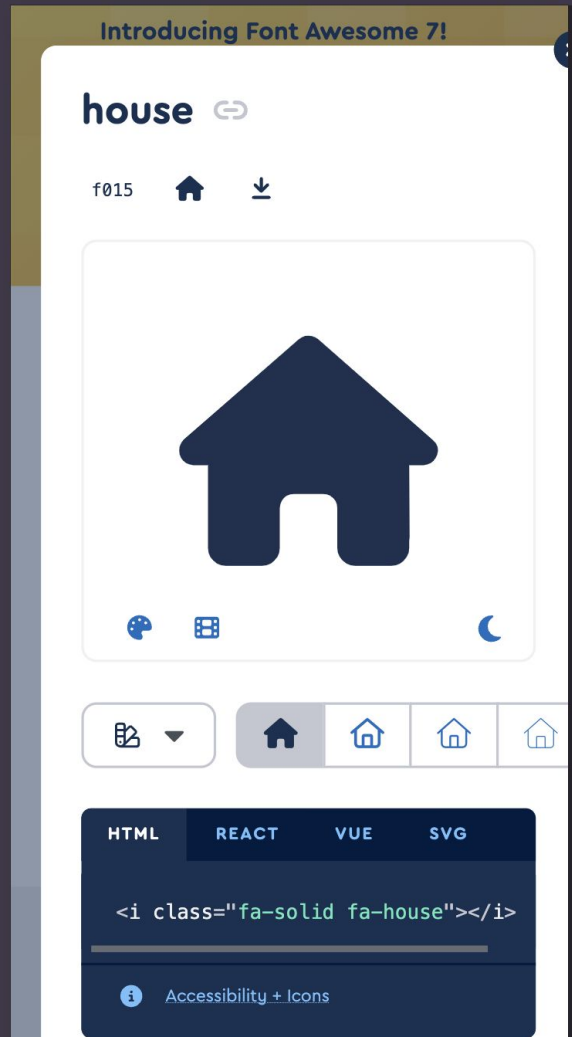
.fa-classic, .fa-light, .fa-
regular, .fa-solid, .fa-thin, .fal,
.far, .fas, .fat {
  font-family: "Font Awesome 6 Pro";
}

.fa, .fa-brands, .fa-classic, .fa-
duotone, .fa-light, .fa-regular,
.fas, .fa-sharp, .fa-sharp-duotone, .fa-solid, .fa-
thin, .fab, .fad, .fadl, .fadr, .fadt, .fal,
.far, .fas, .fasdl, .fasdr, .fasds, .fasdt,
.fasl, .fasr, .fass, .fast, .fat {
  -moz-osx-font-smoothing: grayscale;
  -webkit-font-smoothing: antialiased;
  display: var(--fa-display, inline-block);
  font-style: normal;
  font-variant: normal;
  line-height: 1;
  text-rendering: auto;
}

i, .i, em, .em {
  --font-style: italic;
  font-style: italic;
  font-style: var(--font-style);
}

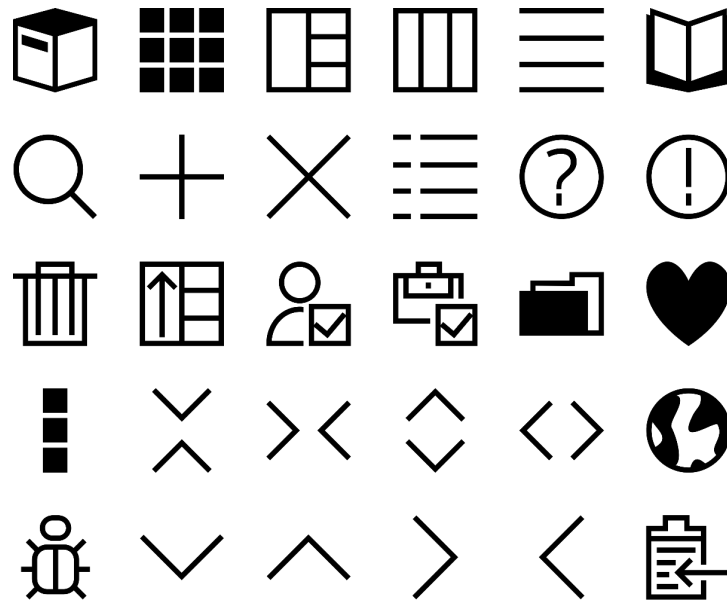
*, ::before, ::after {
  box-sizing: border-box;
}

i {
  font-style: italic;
}
```



The Modern Solution — SVG Sprites

- Vector graphics (sharp at any size)
- One HTTP request
- Style with CSS (color, size, animations!)
- Easily accessible
- Light and fast!

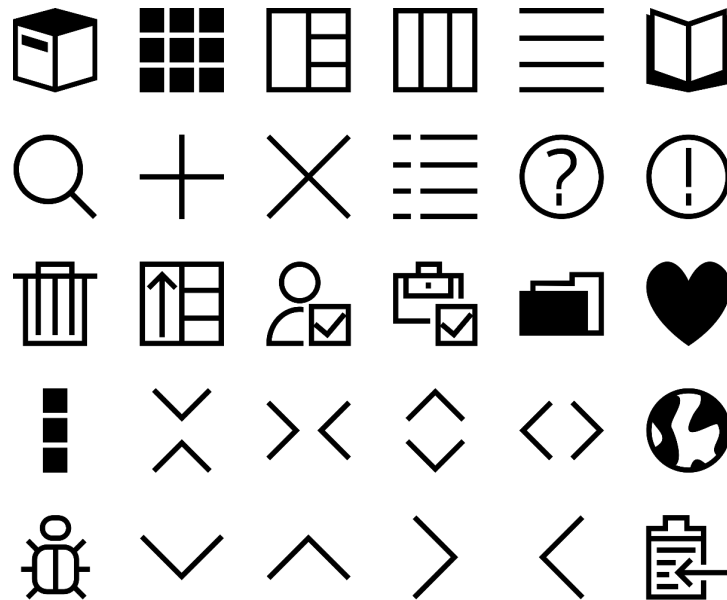


How SVG Sprites Work

SVG sprite = one file with many symbols.

Each icon is a `<symbol>` with an id.

Use `<use>` to reference it anywhere on the page.

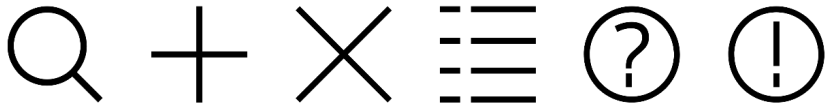


```
<svg viewBox="0 0 100 100" fill="none"
xmlns="http://www.w3.org/2000/svg">
```

```
<symbol id="box">
  // ...
</symbol>
```



```
<symbol id="search">
  // ...
</symbol>
```



```
</svg>
```

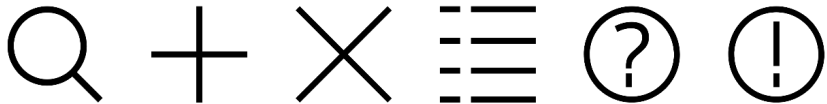
```
<svg class="icon">
  <use xlink:href="./sprite.svg#box"></use>
</svg>
```

```
<svg viewBox="0 0 100 100" fill="none"
xmlns="http://www.w3.org/2000/svg">
```

```
<symbol id="box">
  // ...
</symbol>
```



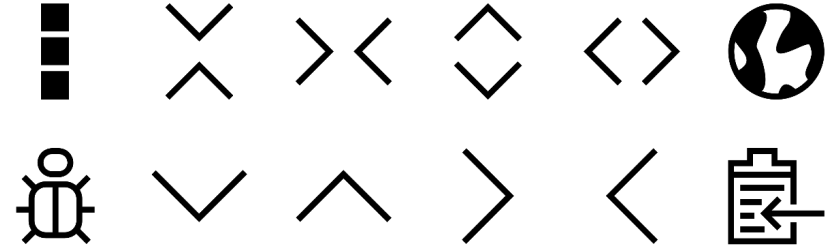
```
<symbol id="search">
  // ...
</symbol>
```



```
<symbol id="bin">
  // ...
</symbol>
```



```
<symbol id="bar">
  // ...
</symbol>
```



```
</svg>
```

New way: symbols + use

✨ best approach

```
<svg viewBox="0 0 100 100" fill="none" xmlns="http://www.w3.org/2000/svg">  
  <symbol id="heart">  
    // ...  
  </symbol>  
  <symbol id="star">  
    // ...  
  </symbol>  
</svg>
```

```
<svg class="icon">  
  <use xlink:href="./sprite.svg#heart"></use>  
</svg>
```

```
<svg class="icon">  
  <use href="./sprite.svg#heart"></use>  
</svg>
```

TWIG

icon.twig

```
{% macro icon(classes, type, width = "24", height = "24") %}  
    <svg class="icon {{ classes }}" width="{{ width }}"  
height="{{ height }}">  
        <use href="/src/assets/sprite.svg#icon-{{ type }}" />  
    </svg>  
{% endmacro %}
```

TWIG

page-header.twig

```
{% from 'ui/icon.twig' import icon %}
```

```
<a href="index.html" class="logo logo--header">
```

```
    {{ icon('logo__image-text', 'logo-text', 134, 21) }}
```

```
</a>
```

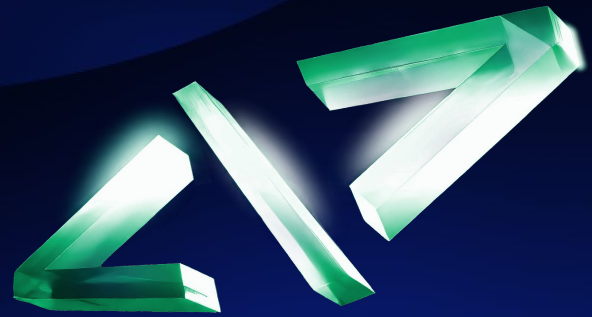
```
<a class="button button--primary page-header__button" href="#">
```

```
    {{ icon('button__icon', 'user') }}
```

```
    Connect
```

```
</a>
```

Styling



Change icon colors on hover

```
.icon {  
    fill: blue;  
    transition: fill 0.3s, stroke 0.3s;  
}
```

```
.icon:hover {  
    fill: red;  
}
```

color: currentColor

[example](#) →

replace colors inside of SVGs in sprite folder to currentColor

```
<svg viewBox="0 0 512 512" class="logo__icon">  
  <path fill="currentColor" d="m48 300 180 169a41 41  
0 0 0 56 0l180-169c31-28 48-68 48-109v-6A143 143 0 0 0  
268 84l-12 12-12-12A143 143 0 0 0 0 185v6c0 41 17 81 48  
109z" />  
</svg>
```

Custom properties

[example](#) →

replace colors inside of SVGs in sprite folder to custom properties with fallback color

```
<svg viewBox="0 0 512 512" class="logo__icon">  
  <path fill="var(--icon-color, blue)" d="m48 300 180  
169a41 41 0 0 0 56 0l180-169c31-28 48-68 48-109v-6A143 143  
0 0 0 268 84l-12 12-12-12A143 143 0 0 0 0 185v6c0 41 17 81  
48 109z" />  
</svg>
```

Custom properties +currentColor

replace colors inside of SVGs in sprite folder to custom properties with fallback color as a currentColor

```
<svg viewBox="0 0 512 512" class="logo__icon">
  <path fill="var(--icon-color, currentColor)" d="m48
300 180 169a41 41 0 0 0 56 0l180-169c31-28 48-68
48-109v-6A143 143 0 0 0 268 84l-12 12-12-12-12A143 143 0 0 0
0 185v6c0 41 17 81 48 109z"/>
</svg>
```

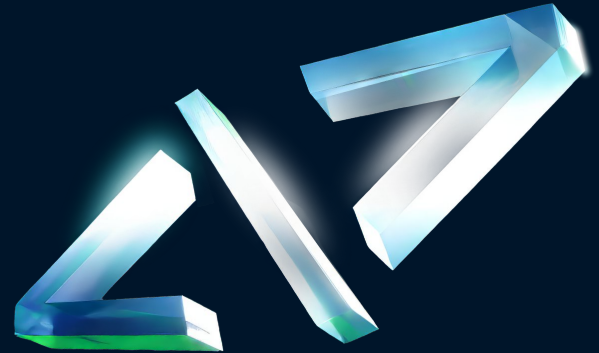
Our the build

let's try to put some icons into sprite folder

`/src/assets/icons`

Homework

1. Put all sprite vector images into the **/src/assets/icons** folder.
2. Use them on pages from sprite via svg use tag.
3. Implement color changes on hover and focus using only CSS for icons that need interactive effects.



Quality Criteria for HTML Course

❤️ Mandatory for passing the course

💛 Required for the highest grade

💚 Optional

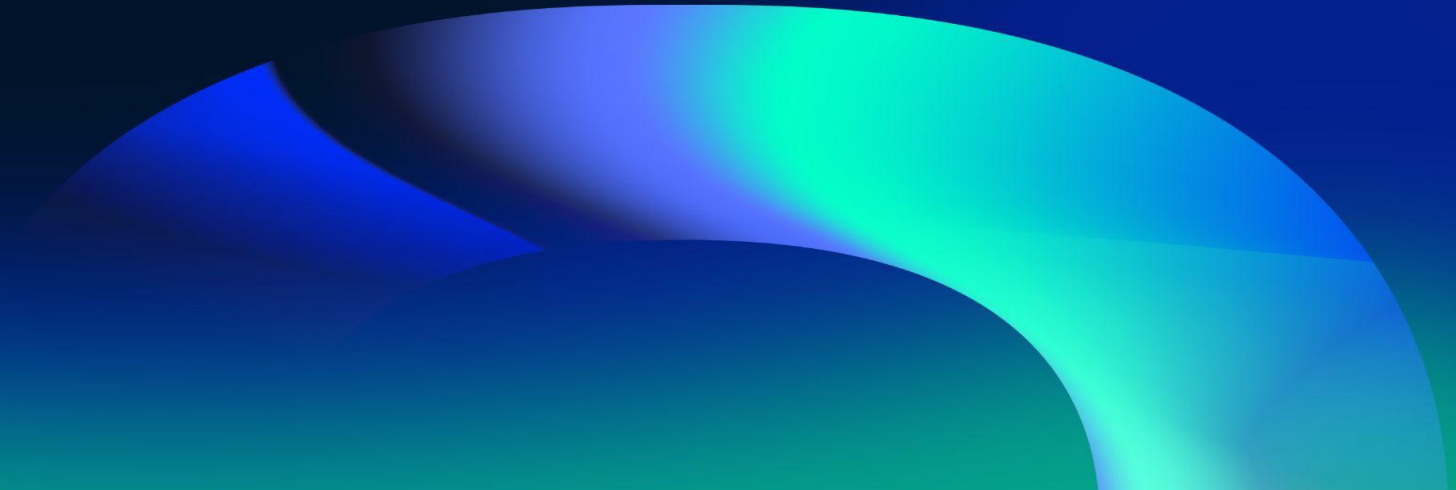
❤️ Use SVG sprite with symbols

B Academy
RO



QUESTIONS?

Please fill out the feedback form
It's very important for us





THANK YOU!

Have a good evening!